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ARTIFICIAL EPIGLOTTIS, AND SWALLOWING AND BREATHING SWITCHING DEVICE

Abstract

An artificial epiglottis is provided that includes a membrane-like piezoelectric element that bends depending on a voltage; and a first driving electrode that applies the voltage to the piezoelectric element. The piezoelectric element is structurally configured to be disposed on a larynx. Based on the applied voltage, the piezoelectric element bends to either a first shape in which an entrance of a trachea is blocked or to a second shape in which the entrance of the trachea is opened.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/JP2023/042888, filed Nov. 30, 2023, which claims priority to Japanese Patent Application No. 2022-192620, filed Dec. 1, 2022, the contents of each of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to an artificial epiglottis and an artificial technique of switching between swallowing and breathing using the artificial epiglottis.

BACKGROUND

[0003] The number of patients with swallowing dysfunction is increasing due to the super-aging society. Swallowing dysfunction may cause aspiration. Aspiration is a phenomenon in which something that should enter the esophagus from the mouth enters the respiratory tract and has various adverse effects on the human body.

[0004] Japanese Unexamined Patent Application Publication No. 2020-54528 describes a swallowing movement assisting orthosis. The orthosis described therein assists the thyroid cartilage to be mechanically pushed upward from a front-lower position in the laryngeal elevation movement that occurs during the swallowing movement pharyngeal stage.

[0005] Moreover, Japanese Unexamined Patent Application Publication No. 11-500339 describes a method for treating dysphagia by electrical stimulation. The method described therein stimulates muscles by electrical stimulation to assist swallowing.

[0006] However, the conventional techniques described above provide swallowing assistance and training. Therefore, it is difficult to sufficiently suppress aspiration with conventional techniques described therein.

SUMMARY OF THE INVENTION

[0007] In view of the foregoing, it is an object of the present disclosure to provide a device that is configured to suppress aspiration more reliably.

[0008] In particular, an artificial epiglottis according to an exemplary embodiment is provided that includes a membrane-like piezoelectric element configured to bend depending on a voltage; and a driving electrode configured to apply the voltage to the piezoelectric element. The piezoelectric element is structurally configured to be disposed on a larynx. The piezoelectric element is, based on the voltage applied by the driving electrode, configured to become a first shape in which an entrance of a trachea is blocked and or a second shape in which the entrance of the trachea is opened.

[0009] With such a configuration, the artificial epiglottis is provided that is configured to switch between a state in which the entrance of the trachea is blocked and a state in which the entrance of the trachea is open by using the bending of the piezoelectric element. Since the artificial epiglottis is formed by the piezoelectric element, switching speed is fast, switching accuracy is high, and posture holding ability of each state is high. Therefore, swallowing and breathing can be switched more reliably.

[0010] According to the exemplary aspects of the present disclosure, aspiration is suppressed more reliably by the artificial epiglottis disclosed described herein.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is an external perspective view showing an example of a configuration of a swallowing and breathing switching device according to a first exemplary embodiment.

[0012] FIGS. 2(A) and 2(B) are views showing a configuration of the swallowing and breathing switching device according to the first exemplary embodiment.

[0013] FIG. 3(A) is a schematic view showing a breathing state in which an artificial epiglottis and the swallowing and breathing switching device according to the first exemplary embodiment are worn, and FIG. 3(B) is a schematic view showing a swallowing state in which the artificial epiglottis and the swallowing and breathing switching device according to the first exemplary embodiment are worn.

[0014] FIGS. 4(A) and 4(B) are views showing a configuration of a swallowing and breathing switching device according to a second exemplary embodiment.

[0015] FIG. 5 is a view showing a configuration of a swallowing and breathing switching device according to a third exemplary embodiment.

[0016] FIG. 6 is a view showing a configuration of a swallowing and breathing switching device according to a fourth exemplary embodiment.

[0017] FIG. 7 is a view showing a configuration of a swallowing and breathing switching device according to a fifth exemplary embodiment.

[0018] FIG. 8 is a view showing a configuration of a swallowing and breathing switching device according to a sixth exemplary embodiment.

[0019] FIG. 9 is a view showing a configuration of a swallowing and breathing switching device according to a seventh exemplary embodiment.

[0020] FIG. 10 is a view showing a configuration of a swallowing and breathing switching device according to an eighth exemplary embodiment.

[0021] FIG. 11 is a view showing a configuration of a swallowing and breathing switching device according to a ninth exemplary embodiment.

[0022] FIG. 12(A), FIG. 12(B), and FIG. 12(C) are perspective views each showing a configuration example of an artificial epiglottis according to a tenth exemplary embodiment.

[0023] FIG. 13 is a schematic view showing a curved state of the artificial epiglottis shown in FIG. 12(A).

[0024] FIG. 14 is a view showing a configuration of a swallowing and breathing switching device according to an eleventh exemplary embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

First Exemplary Embodiment

[0025] An artificial epiglottis and a swallowing and breathing switching device according to a first exemplary embodiment will be described with reference to the drawings. FIG. 1 is an external perspective view of the swallowing and breathing switching device according to the first exemplary embodiment. FIGS. 2(A) and 2(B) are views showing a configuration of the swallowing and breathing switching device according to the first exemplary embodiment, wherein FIG. 2(A) shows a first shape of the artificial epiglottis and FIG. 2(B) shows a second shape of the artificial epiglottis. FIG. 3(A) is a schematic view showing a breathing state in which the artificial epiglottis and the swallowing and breathing switching device according to the first exemplary embodiment are worn, and FIG. 3(B) is a schematic view showing a swallowing state in which the artificial epiglottis and the swallowing and breathing switching device according to the first exemplary embodiment are worn.

Physical Configuration of Swallowing and Breathing Switching Device 10

[0026] As shown in FIGS. 1, 2(A) and 2(B), a swallowing and breathing switching device 10

includes an artificial epiglottis **20**, a drive signal generating unit **31**, an orthosis **32**, a physical switch **39**, and a cable **40**.

[0027] The drive signal generating unit **31** is configured by using at least one of an electric circuit element and an electronic circuit element. The drive signal generating unit **31** is incorporated in a predetermined housing **H31**.

[0028] The physical switch **39** is provided in the housing **H31**. The physical switch **39** is electrically connected to the drive signal generating unit **31**.

[0029] The orthosis **32** has a belt-like shape. The housing **H31** is fixed to the front side of the orthosis **32**.

[0030] The cable **40** has a first cable **41**, a second cable **42**, and a third cable **43**. The first cable **41**, the second cable **42**, and the third cable **43** are each a conductive wire whose peripheral surface is covered with an insulating material, for example. The insulating material of the first cable **41**, the second cable **42**, and the third cable **43** may be a biocompatible material in an exemplary aspect.

[0031] As shown, one end of the cable **40** (the first cable **41**, the second cable **42**, and the third cable **43**) is connected to the drive signal generating unit **31**. The cable **40** (the first cable **41**, the second cable **42**, and the third cable **43**) is disposed so as to extend from the back surface of the orthosis **32**. The other end of the cable **40** (the first cable **41**, the second cable **42**, and the third cable **43**) is connected to the artificial epiglottis **20**.

[0032] The artificial epiglottis **20** includes a piezoelectric element **210**, a first driving electrode **221**, a second driving electrode **222**, and a third driving electrode **223**. The piezoelectric element **210** includes a first piezoelectric element **211** and a second piezoelectric element **212**.

[0033] The first piezoelectric element **211** and the second piezoelectric element **212** have a flat membrane-like shape. The main material of the first piezoelectric element **211** and the second piezoelectric element **212** is, for example, polylactic acid. The first piezoelectric element **211** and the second piezoelectric element **212** are configured to extend or contract in a direction parallel to the flat membrane surfaces thereof depending on the voltage applied thereto.

[0034] In the exemplary aspect, the first piezoelectric element **211** and the second piezoelectric element **212** are laminated together so that the flat membrane surfaces thereof are parallel to each other.

[0035] The first driving electrode **221** is disposed on the flat membrane surface of the first piezoelectric element **211** on a side opposite to the surface facing the second piezoelectric element **212**. The second driving electrode **222** is disposed on the flat membrane surface of the second piezoelectric element **212** on a side opposite to the surface facing the first piezoelectric element **211**. The third driving electrode **223** is disposed between the first piezoelectric element **211** and the second piezoelectric element **212**. Thus, the first piezoelectric element **211** is sandwiched between the first driving electrode **221** and the third driving electrode **223**. The second piezoelectric element **212** is sandwiched between the second driving electrode **222** and the third driving electrode **223**.

[0036] In the artificial epiglottis **20** having this configuration according to the exemplary aspect, the first piezoelectric element **211** does not expand or contract when the third driving electrode **223** and the first driving electrode **221** have the same potential. Similarly, the second piezoelectric element **212** does not expand or contract when the third driving electrode **223** and the second driving electrode **222** have the same potential. Therefore, the artificial epiglottis **20** is not bent (e.g., curved) as shown in FIG. 2(A). This state is the first shape of the artificial epiglottis **20**.

[0037] On the other hand, when a voltage is applied to the first piezoelectric element **211** so that the first driving electrode **221** has a positive potential with respect to the third driving electrode **223**, the first piezoelectric element **211** will extend in a direction parallel to the flat membrane surface. When a voltage is applied to the second piezoelectric element **212** so that the second driving electrode **222** has a positive potential with respect to the third driving electrode **223**, the second piezoelectric element **212** will contract in a direction parallel to the flat membrane surface.

[0038] In the exemplary aspect, the first piezoelectric element **211** and the second piezoelectric

element **212** are laminated together in a direction orthogonal to the flat membrane surface. As described above, when the first piezoelectric element **211** extends and the second piezoelectric element **212** contracts, the artificial epiglottis **20** composed of a laminated structure of the first piezoelectric element **211** and the second piezoelectric element **212** bends (e.g., curves) so as to bend toward the second piezoelectric element **212** as shown in FIG. 2(B). This state is the second shape of the artificial epiglottis **20**. At this time, the extension due to the voltage applied to the first piezoelectric element **211** and the contraction due to the voltage applied to the second piezoelectric element **212** are synergized, so that the artificial epiglottis **20** bends larger than a case in which only the first piezoelectric element **211** or only the second piezoelectric element **212** is provided.

[0039] According to an exemplary aspect, the first

[0040] piezoelectric element **211**, the second piezoelectric element **212**, the first driving electrode **221**, the second driving electrode **222**, and the third driving electrode **223** forming the artificial epiglottis **20** are preferably made of a biocompatible material. However, as shown in FIG. 1, the artificial epiglottis **20** may include a cover **C20**. In this regard, the cover **C20** is flexible enough not to inhibit the bending of the artificial epiglottis **20**. When the cover **C20** is provided, the first piezoelectric element **211**, the second piezoelectric element **212**, the first driving electrode **221**, the second driving electrode **222**, and the third driving electrode **223** do not have to be made of a biocompatible material as long as the cover **C20** is made of a biocompatible material.

Electrical Configuration of swallowing and Breathing Switching Device **10**

[0041] The drive signal generating unit **31** of the swallowing and breathing switching device **10** includes a power supply circuit **311**, an electric switch **312**, an electric switch **313**, and a switch control circuit **310**.

[0042] The power supply circuit **311** includes a direct-current power supply **DC1** and a direct-current power supply **DC2**. The direct-current power supply **DC1** and the direct-current power supply **DC2** are, for example, primary batteries or secondary batteries.

[0043] The negative electrode of the direct-current power supply **DC1** and the negative electrode of the direct-current power supply **DC2** are connected to each other. The node of the negative electrode of the direct-current power supply **DC1** and the negative electrode of the direct-current power supply **DC2** are connected to the third driving electrode **223** of the artificial epiglottis **20** through the third cable **43**.

[0044] As shown, the positive electrode of the direct-current power supply **DC1** is connected to the electric switch **312**. The electric switch **312** is connected to the first driving electrode **221** of the artificial epiglottis **20** through the first cable **41**.

[0045] The positive electrode of the direct-current power supply **DC2** is connected to the electric switch **313**. The electric switch **313** is connected to the second driving electrode **222** of the artificial epiglottis **20** through the second cable **42**.

[0046] The switch control circuit **310** is connected to the physical switch **39** and connected to the electric switch **312** and the electric switch **313**.

[0047] The switch control circuit **310** generates a switch control signal for the electric switch **312** and the electric switch **313** according to the operating state of the physical switch **39**. The electric switch **312** and the electric switch **313** are configured to switch between open and short-circuit according to the switch control signal.

[0048] For example, in a state in which the physical switch **39** is not operated, the switch control circuit **310** does not output the switch control signal to the electric switch **312** and the electric switch **313**. In such a case, the electric switch **312** and the electric switch **313** are in an open state. Therefore, no direct-current voltage for driving is applied to the artificial epiglottis **20**. As a result, the artificial epiglottis **20** becomes the first shape shown in FIG. 2(A).

[0049] On the other hand, in a state in which the physical switch **39** is operated, the switch control circuit **310** is configured to generate a switch control signal for switching between a Hi state and a Low state at a predetermined period, and to output the generated switch control signal to the

electric switch **312** and the electric switch **313**.

[0050] The electric switch **312** and the electric switch **313** are in a short-circuit state when the switch control signal is in the Hi state and are in an open state when the switch control signal is in the Low state.

[0051] When the electric switch **312** and the electric switch **313** are in the open state, no direct-current voltage for driving is applied to the artificial epiglottis **20** (an example of a first state of the drive signal). Therefore, the artificial epiglottis **20** becomes the first shape shown in FIG. 2(A).

[0052] When the electric switch **312** and the electric switch **313** are in the short-circuit state, a direct-current voltage for driving is applied to the artificial epiglottis **20** (an example of a second state of the drive signal). Therefore, the artificial epiglottis **20** becomes the second shape shown in FIG. 2(B).

[0053] Thus, the swallowing and breathing switching device **10** can be configured to switch between the first shape in which the artificial epiglottis **20** is not bent and the second shape in which the artificial epiglottis **20** is bent.

Wearing on Human Body and Practical Wearing Condition

[0054] As shown in FIGS. 3(A) and 3(B), the artificial epiglottis **20** is disposed on the larynx of a body. More specifically, one end of the artificial epiglottis **20** in the direction in which the first piezoelectric element **211** and the second piezoelectric element **212** expand and contract is fixed to the larynx. The other end of the artificial epiglottis **20** in the direction in which the first piezoelectric element **211** and the second piezoelectric element **212** expand and contract is a free end not fixed to the larynx.

[0055] In a state in which no voltage is applied to the artificial epiglottis **20**, the artificial epiglottis **20** is disposed so as not to block the entrance of the trachea, as shown in FIG. 3(A). Further, in a state in which a voltage is applied to the artificial epiglottis **20**, the artificial epiglottis **20** is disposed so as to block the entrance of the trachea, as shown in FIG. 3(B).

[0056] The housing **H31**, in which the drive signal generating unit **31** is incorporated, is worn on the front neck of the body by the orthosis **32**. The cable **40** (the first cable **41**, the second cable **42**, and the third cable **43**) is disposed through the larynx, and connects the drive signal generating unit **31** and the artificial epiglottis **20**.

[0057] In such a state, when the wearer presses the physical switch **39**, the switch control circuit **310** is driven. The switch control circuit **310** is configured to switch between opening and short-circuiting of the electric switch **312** and the electric switch **313**.

[0058] More specifically, when the drive signal generating unit **31** (the switch control circuit **310**) opens the electric switch **312** and the electric switch **313**, the artificial epiglottis **20** is configured to become (e.g., change or bend) the first shape, so that the entrance of the trachea is not blocked, as shown in FIG. 3(A). Therefore, air from the nasal cavity enters the trachea, so that breathing is performed without any problem.

[0059] On the other hand, when the drive signal generating unit **31** (the switch control circuit **310**) short-circuits the electric switch **312** and the electric switch **313**, the artificial epiglottis **20** is configured to become (e.g., change or bend) the second shape, so that the entrance of the trachea is blocked by the artificial epiglottis **20**, as shown in FIG. 3(B). Therefore, food (drink) from the oral cavity enters the esophagus, so that swallowing is performed without any problem.

[0060] The above configuration shows an aspect of switching between the opening and short-circuiting of the electric switch **312** and the electric switch **313**, in other words, between the first shape and the second shape of the artificial epiglottis **20**, at a predetermined period. However, the swallowing and breathing switching device **10** may selectively open and short-circuit the electric switch **312** and the electric switch **313**. Specifically, the swallowing and breathing switching device **10** can be configured to perform control to short-circuit the electric switch **312** and the electric switch **313** only when swallowing is performed. The swallowing and breathing switching device **10** can be configured to perform control to open the electric switch **312** and the electric switch **313**

when swallowing is not performed (i.e., when breathing).

[0061] As described above, by providing the artificial epiglottis **20** and the swallowing and breathing switching device **10**, aspiration is suppressed more reliably.

[0062] Since aspiration is suppressed more reliably, aspiration pneumonia is also suppressed. Further, since aspiration is suppressed more reliably, food can be taken from the mouth without the need for a gastrostomy or enterostomy. Further, since aspiration can be suppressed more reliably, a tracheostomy is not required. Further, since aspiration is suppressed more reliably, food restrictions such as having to make food thick can be eliminated.

[0063] By providing the artificial epiglottis **20** and the swallowing and breathing switching device **10**, tracheal cannula-related problems (such as pain during regular replacement and choking due to blockage in the tracheal cannula) can be prevented. Further, by providing the artificial epiglottis **20** and the swallowing and breathing switching device **10**, rehabilitation to train the periphery of the larynx is not required.

[0064] Further, in the swallowing and breathing switching device **10**, the first piezoelectric element **211** and the second piezoelectric element **212** are used in the artificial epiglottis **20**. With such a configuration, the swallowing and breathing switching device **10** can be configured to switch (e.g., adapted or bent) between the first shape and the second shape of the artificial epiglottis **20** at high speed. In addition, the artificial epiglottis **20** can stably hold the first shape and the second shape. In addition, the artificial epiglottis **20** can be realized with low power consumption. In addition, electromagnetic noise generated by the artificial epiglottis **20** is suppressed. In addition, the artificial epiglottis **20** can be formed in a small size and thin shape so as to be easily disposed at a desired position of the larynx. In addition, the artificial epiglottis **20** can be realized with a simple configuration.

[0065] As shown in FIG. **1(A)**, for example, the artificial epiglottis **20** has two piezoelectric elements (the first piezoelectric element **211** and the second piezoelectric element **212**) laminated together. Further, the first piezoelectric element **211** and the second piezoelectric element **212** have opposite expansion and contraction directions (e.g., displacement directions) when a voltage is applied. Therefore, the bending effect caused by the first piezoelectric element **211** and the bending effect caused by the second piezoelectric element **212** are synergized. Thus, the artificial epiglottis **20** can realize a larger bending amount. As a result, the artificial epiglottis **20** can increase the difference between the first shape in which the entrance of the trachea is not blocked and the second shape in which the entrance of the trachea is blocked, so that aspiration is suppressed more reliably.

[0066] In the present embodiment, an aspect has been described in which the piezoelectric element **210** uses a multilayer body of the first piezoelectric element **211** and the second piezoelectric element **212** (e.g., a bimorph drive). However, the piezoelectric element **210** may alternatively be provided with at least one of the first piezoelectric element **211** and the second piezoelectric element **212** (e.g., an unimorph drive).

Second Exemplary Embodiment

[0067] An artificial epiglottis and a swallowing and breathing switching device according to a second exemplary embodiment will be described with reference to the drawings. FIGS. **4(A)** and **4(B)** are views showing a configuration of a swallowing and breathing switching device according to a second exemplary embodiment, wherein FIG. **4(A)** shows a first shape of the artificial epiglottis and FIG. **4(B)** shows a second shape of the artificial epiglottis.

[0068] As shown in FIGS. **4(A)** and **4(B)**, a swallowing and breathing switching device **10A** according to the second embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10A** includes an artificial epiglottis **20A** and a drive signal generating unit **31A**. The other configurations of the swallowing and breathing switching device **10A** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0069] The swallowing and breathing switching device **10A** includes the artificial epiglottis **20A**, the drive signal generating unit **31A**, and a cable **40A**. The cable **40A** includes a first cable **41** and a second cable **42**.

[0070] The artificial epiglottis **20A** includes a first piezoelectric element **211**, a second piezoelectric element **212**, a first driving electrode **221**, and a second driving electrode **222**. The first piezoelectric element **211** and the second piezoelectric element **212** are laminated in contact with each other. The first driving electrode **221** is disposed on a surface of the first piezoelectric element **211** on a side opposite to the contact surface of the second piezoelectric element **212**. The second driving electrode **222** is disposed on a surface of the second piezoelectric element **212** on a side opposite to the contact surface of the first piezoelectric element **211**.

[0071] When a voltage is applied to the first piezoelectric element **211** so that the first driving electrode **221** has a positive potential with respect to the second driving electrode **222**, the first piezoelectric element **211** will extend in a direction parallel to the flat membrane surface. When a voltage is applied to the second piezoelectric element **212** so that the first driving electrode **221** has a positive potential with respect to the second driving electrode **222**, the second piezoelectric element **212** will contract in a direction parallel to the flat membrane surface.

[0072] The drive signal generating unit **31A** includes a power supply circuit **311A**, an electric switch **312**, and a switch control circuit **310**. The power supply circuit **311A** includes a direct-current power supply **DC10**. The positive electrode of the direct-current power supply **DC10** is connected to the first driving electrode **221** through the electric switch **312** and the first cable **41**. The negative electrode of the direct-current power supply **DC10** is connected to the second driving electrode **222** through the second cable **42**.

[0073] With such a configuration, when the electric switch **312** is short-circuited, a voltage from the direct-current power supply **DC10** is applied to the artificial epiglottis **20A** (an example of a first state of the drive signal). When the electric switch **312** is opened, no voltage is applied to the artificial epiglottis **20A** (an example of a second state of the drive signal).

[0074] With such a configuration, the swallowing and breathing switching device **10A** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and breathing switching device **10**, so that aspiration is suppressed more reliably.

Third Exemplary Embodiment

[0075] An artificial epiglottis and a swallowing and breathing switching device according to the third exemplary embodiment will be described with reference to the drawings. FIG. 5 is a view showing a configuration of the swallowing and breathing switching device according to the third exemplary embodiment.

[0076] As shown in FIG. 5, a swallowing and breathing switching device **10B** according to the third embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10B** includes an artificial epiglottis **20B**, a drive signal generating unit **31B**, and a cable **40B**. The artificial epiglottis **20B** is the same as the artificial epiglottis **20B** according to the second embodiment, and the cable **40B** is the same as the cable **40A** according to the second embodiment.

[0077] The drive signal generating unit **31B** includes a power supply circuit **311B**. The power supply circuit **311B** includes an alternating-current power supply **AC**. One output terminal of the alternating-current power supply **AC** is connected to the first driving electrode **221** through the first cable **41**. The other output terminal of the alternating-current power supply **AC** is connected to the second driving electrode **222** through the second cable **42**. The physical switch **39** is connected to the power supply circuit **311B**.

[0078] The power supply circuit **311B** receives an operation signal from the physical switch **39** and drives the alternating-current power supply **AC**. The alternating-current power supply **AC** applies an alternating-current voltage to the artificial epiglottis **20B**. The alternating-current voltage is a

voltage that alternates between a Hi state (an example of a first state of the drive signal) and a Low state (an example of a second state of the drive signal).

[0079] For example, the artificial epiglottis **20B** becomes a first shape in which it does not bend when the alternating-current voltage is in the Low state and becomes a second shape in which it bends when the alternating-current voltage is in the Hi state.

[0080] For example, the period of the alternating-current

[0081] voltage corresponds to the period of salivation during sleep, the period of breathing and/or the like.

[0082] With such a configuration, the swallowing and breathing switching device **10B** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and breathing switching device **10**, so that aspiration is suppressed more reliably.

Fourth Exemplary Embodiment

[0083] An artificial epiglottis and a swallowing and breathing switching device according to a fourth exemplary embodiment will be described with reference to FIG. **6**. FIG. **6** is a view showing a configuration of the swallowing and breathing switching device according to the fourth exemplary embodiment.

[0084] As shown in FIG. **6**, a swallowing and breathing switching device **10C** according to the fourth embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10C** includes a drive signal generating unit **31C**. The other configurations of the swallowing and breathing switching device **10C** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0085] The swallowing and breathing switching device **10C** includes the drive signal generating unit **31C**. The drive signal generating unit **31C** includes a power supply circuit **311C**, a power transmission control unit **314**, a power transmission coil **315**, a power reception coil **316**, and a power reception control unit **317**. The power supply circuit **311C**, the power transmission control unit **314**, and the power transmission coil **315** are disposed outside the body. The power reception coil **316** and the power reception control unit **317** are disposed inside the body.

[0086] The power supply circuit **311C** includes a direct-current power supply **DC10**. The direct-current power supply **DC10** is connected to the power transmission control unit **314**. The power transmission control unit **314** is connected to the power transmission coil **315**.

[0087] The power reception coil **316** is connected to the power reception control unit **317**. The power reception control unit **317** is connected to the artificial epiglottis **20** through a cable **40** (a first cable **41**, a second cable **42**, and a third cable **43**).

[0088] The power transmission control unit **314** receives an operation signal from the physical switch **39**, converts a direct-current voltage from the direct-current power supply **DC10**, and supplies an alternating-current power transmission current to the power transmission coil **315**. The power transmission coil **315** is configured to excite an alternating magnetic field by the alternating-current power transmission current.

[0089] The power reception coil **316** is coupled to the alternating magnetic field to generate an alternating-current power reception current. The power reception control unit **317** rectifies the power reception current, generates a direct-current voltage, and supplies the direct-current voltage to the artificial epiglottis **20**.

[0090] As described above, the drive signal generating unit **31C** employs a wireless power supply system.

[0091] With such a configuration, the swallowing and breathing switching device **10C** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and breathing switching device **10**, so that aspiration is suppressed more reliably.

[0092] Note that the power reception control unit **317** may include a storage battery. By charging the storage battery, the swallowing and breathing switching device **10C** can supply a voltage to the artificial epiglottis **20** even in a state in which no unit is provided outside the body in the drive signal generating unit **31C**.

Fifth Exemplary Embodiment

[0093] An artificial epiglottis and a swallowing and breathing switching device according to a fifth exemplary embodiment will be described with reference to the drawings. FIG. **7** is a view showing a configuration of the swallowing and breathing switching device according to the fifth exemplary embodiment.

[0094] As shown in FIG. **7**, a swallowing and breathing switching device **10D** according to the fifth embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10D** includes a drive signal generating unit **31D**. The other configurations of the swallowing and breathing switching device **10D** are the same as those of the swallowing and breathing switching device **10**, and description of the same parts will be omitted.

[0095] The swallowing and breathing switching device **10D** includes the drive signal generating unit **31D**. The drive signal generating unit **31D** is different from the drive signal generating unit **31** in that the switch control circuit **310** is omitted.

[0096] In the drive signal generating unit **31D**, the electric switch **312** and the electric switch **313** are directly connected to the physical switch **39**. The electric switch **312** and the electric switch **313** switch between an open state and a short-circuit state in response to an operation signal from the physical switch **39**.

[0097] With this configuration, the swallowing and breathing switching device **10D** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked by directly reflecting the operation of the wearer.

Sixth Exemplary Embodiment

[0098] An artificial epiglottis and a swallowing and breathing switching device according to a sixth exemplary embodiment will be described with reference to the drawings. FIG. **8** is a view showing a configuration of the swallowing and breathing switching device according to the sixth exemplary embodiment.

[0099] As shown in FIG. **8**, a swallowing and breathing switching device **10E** according to the sixth embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that, in the swallowing and breathing switching device **10E**, the physical switch **39** is omitted and a myoelectric sensor **51** and a control signal generating unit **52** are provided. The other configurations of the swallowing and breathing switching device **10E** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0100] The swallowing and breathing switching device **10E** includes the myoelectric sensor **51** and the control signal generating unit **52**. The myoelectric sensor **51** is disposed in the oral cavity of the body on which the artificial epiglottis **20** is worn. The myoelectric sensor **51** detects the movement of the muscles of the oral cavity and generates a myoelectric detection signal. The myoelectric sensor **51** outputs the myoelectric detection signal to the control signal generating unit **52**.

[0101] The control signal generating unit **52** stores the myoelectric detection signal during swallowing and the myoelectric detection signal during breathing in advance in a memory. The control signal generating unit **52** may alternatively store at least the myoelectric detection signal during swallowing in the memory.

[0102] If the myoelectric detection signal from the myoelectric sensor **51** and the myoelectric detection signal during swallowing are the same, the control signal generating unit **52** outputs a control signal indicating a swallowing state to the switch control circuit **310**. When the control signal indicating the swallowing state is received, the switch control circuit **310** generates a switch

control signal for performing a control to short-circuit the electric switch **312** and the electric switch **313**. The switch control circuit **310** outputs the switch control signal to the electric switch **312** and the electric switch **313**.

[0103] When the electric switch **312** and the electric switch **313** are short-circuited by the switch control signal, a voltage is applied to the artificial epiglottis **20**. Therefore, the artificial epiglottis **20** becomes a second shape in which the entrance of the trachea is blocked.

[0104] On the other hand, if the myoelectric detection signal from the myoelectric sensor **51** and the myoelectric detection signal during swallowing are not the same, the control signal generating unit **52** does not output the control signal to the switch control circuit **310**. When the control signal is not received from the control signal generating unit **52**, the switch control circuit **310** does not generate a switch control signal for performing a control to short-circuit the electric switch **312** and the electric switch **313**.

[0105] When the electric switch **312** and the electric switch **313** are opened, no voltage is applied to the artificial epiglottis **20**. Therefore, the artificial epiglottis **20** becomes a first shape in which the entrance of the trachea is not blocked.

[0106] With the above-described configuration and control, the swallowing and breathing switching device **10E** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and breathing switching device **10**, so that aspiration is suppressed more reliably.

[0107] Further, the swallowing and breathing switching device **10E** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked in response to the mouth movement of the wearer of the artificial epiglottis **20**. Therefore, the swallowing and breathing switching device **10E** can suppress aspiration more reliably.

Seventh Exemplary Embodiment

[0108] An artificial epiglottis and a swallowing and breathing switching device according to a seventh exemplary embodiment will be described with reference to the drawings. FIG. **9** is a view showing a configuration of the swallowing and breathing switching device according to the seventh exemplary embodiment.

[0109] As shown in FIG. **9**, a swallowing and breathing switching device **10F** according to the seventh embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that, in the swallowing and breathing switching device **10F**, the physical switch **39** is omitted and a brain wave sensor **61** and a control signal generating unit **62** are provided. The other configurations of the swallowing and breathing switching device **10F** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0110] The swallowing and breathing switching device **10F** includes the brain wave sensor **61** and the control signal generating unit **62**. The brain wave sensor **61** is disposed on the head on which the artificial epiglottis **20** is worn. The brain wave sensor **61** detects a brain wave and generates a brain wave detection signal. The brain wave sensor **61** outputs the brain wave detection signal to the control signal generating unit **62**.

[0111] The control signal generating unit **62** stores the brain wave detection signal during swallowing and the brain wave detection signal during breathing in advance in a memory. The control signal generating unit **62** may alternatively store at least the brain wave detection signal during swallowing in the memory.

[0112] If the brain wave detection signal from the brain wave sensor **61** and the brain wave detection signal during swallowing are the same, the control signal generating unit **62** is configured to output a control signal indicating the swallowing state to the switch control circuit **310**. When the control signal indicating the swallowing state is received, the switch control circuit **310** is

configured to generate a switch control signal for performing a control to short-circuit the electric switch **312** and the electric switch **313**. The switch control circuit **310** is then configured to output the switch control signal to the electric switch **312** and the electric switch **313**.

[0113] When the electric switch **312** and the electric switch **313** are short-circuited by the switch control signal, a voltage is applied to the artificial epiglottis **20**. Therefore, the artificial epiglottis **20** becomes a second shape in which the entrance of the trachea is blocked.

[0114] On the other hand, if the myoelectric detection signal from the brain wave sensor **61** and the myoelectric detection signal during swallowing are not the same, the control signal generating unit **62** does not output the control signal to the switch control circuit **310**. When the control signal is not received from the control signal generating unit **62**, the switch control circuit **310** does not generate a switch control signal for performing a control to short-circuit the electric switch **312** and the electric switch **313**.

[0115] When the electric switch **312** and the electric switch **313** are opened, no voltage is applied to the artificial epiglottis **20**. Therefore, the artificial epiglottis **20** becomes a first shape in which the entrance of the trachea is not blocked.

[0116] With the above-described configuration and control, the swallowing and breathing switching device **10F** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and breathing switching device **10**, so that aspiration is suppressed more reliably.

[0117] Further, the swallowing and breathing switching device **10F** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked in response to the brain wave of the wearer of the artificial epiglottis **20** during swallowing and during breathing. Therefore, the swallowing and breathing switching device **10F** suppresses aspiration more reliably.

Eighth Exemplary Embodiment

[0118] An artificial epiglottis and a swallowing and breathing switching device according to an eighth exemplary embodiment will be described with reference to the drawings. FIG. **10** is a view showing a configuration of the swallowing and breathing switching device according to the eighth exemplary embodiment.

[0119] As shown in FIG. **10**, a swallowing and breathing switching device **10G** according to the eighth embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10G** includes a drive signal generating unit **31G**. The other configurations of the swallowing and breathing switching device **10G** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0120] The swallowing and breathing switching device **10G** includes the drive signal generating unit **31G**. The drive signal generating unit **31G** is different from the drive signal generating unit **31** according to the first embodiment in that the drive signal generating unit **31G** includes a power supply circuit **311G** and a drive control circuit **390**. The other configurations of the drive signal generating unit **31G** are the same as those of the drive signal generating unit **31**, and descriptions of the same parts will be omitted.

[0121] The power supply circuit **311G** includes a variable voltage generator **3161**, a variable voltage generator **3162**, and a memory **3163**. In an exemplary aspect, the variable voltage generator **3161** and the variable voltage generator **3162** are programmable variable voltage devices, and the voltage value of the direct-current voltage to be outputted can be set.

[0122] The memory **3163** is configured to store a voltage value corresponding to the wearer. More specifically, the memory **3163** stores the relationship between the bending angle of the artificial epiglottis **20** and the voltage corresponding to the bending angle for the wearer of the artificial epiglottis **20**.

[0123] The drive control circuit **390** is configured to control the variable voltage generator **3161** and the variable voltage generator **3162** based on the relationship between the bending angle of the artificial epiglottis **20** and the voltage corresponding to the bending angle. Based on the control from the drive control circuit **390**, the variable voltage generator **3161** and the variable voltage generator **3162** are configured to generate a direct-current voltage corresponding to the voltage value stored in the memory **3163**.

[0124] The drive control circuit **390** is configured to control the drive of the variable voltage generator **3161** and the variable voltage generator **3162**, and to control the opening and short-circuiting of the electric switch **312** and the electric switch **313**. The control of the electric switch **312** and the electric switch **313** by the drive control circuit **390** is similar to the control of the switch control circuit **310** described above.

[0125] With the above-described configuration and control, the swallowing and breathing switching device **10G** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and breathing switching device **10**, so that aspiration is suppressed more reliably.

[0126] Further, the swallowing and breathing switching device **10G** can be configured to control the voltage value of the voltage applied to the artificial epiglottis **20** by the variable voltage generator **3161** and the variable voltage generator **3162**. Thus, the bending amount of the artificial epiglottis **20** can be adjusted according to the wearer. Therefore, the swallowing and breathing switching device **10G** suppresses aspiration more reliably.

Ninth Exemplary Embodiment

[0127] An artificial epiglottis and a swallowing and breathing switching device according to a ninth exemplary embodiment will be described with reference to the drawings. FIG. **11** is a view showing a configuration of the swallowing and breathing switching device according to the ninth exemplary embodiment.

[0128] As shown in FIG. **11**, a swallowing and breathing switching device **10H** according to the ninth embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10H** includes a connector **318** and a connector **49**. The other configurations of the swallowing and breathing switching device **10H** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0129] The swallowing and breathing switching device **10H** includes a drive signal generating unit **31H**. The drive signal generating unit **31H** is different from the drive signal generating unit **31** according to the first embodiment in that the drive signal generating unit **31H** includes the connector **318**. The other configurations of the drive signal generating unit **31H** are the same as those of the drive signal generating unit **31**, and descriptions of the same parts will be omitted.

[0130] The connector **318** is connected to the electric switch **312**, the electric switch **313**, and the node of the direct-current power supply DC1 and direct-current power supply DC2 in the power supply circuit **311**.

[0131] The connector **49** is connected to a cable **40** (a first cable **41**, a second cable **42**, and a third cable **43**).

[0132] The connector **318** and the connector **49** are detachably connected to each other. When the connector **318** and the connector **49** are connected, the electric switch **312** is connected to the first cable **41**, and the electric switch **313** is connected to the second cable **42**. The node of the direct-current power supply DC1 and direct-current power supply DC2 in the power supply circuit **311** are connected to the third cable **43**.

[0133] With such a configuration, the swallowing and breathing switching device **10H** can be configured to switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked, in the same manner as the swallowing and

breathing switching device **10**, so that aspiration is suppressed more reliably.

[0134] Further, in the swallowing and breathing switching device **10H**, the drive signal generating unit **31H** can be replaced. Therefore, the swallowing and breathing switching device **10H** is more user-friendly for the wearer.

Tenth Exemplary Embodiment

[0135] An artificial epiglottis according to a tenth exemplary embodiment will be described with reference to the drawings. FIG. **12(A)**, FIG. **12(B)**, and FIG. **12(C)** are perspective views each showing a configuration example of the artificial epiglottis according to the tenth exemplary embodiment. FIG. **13** is a schematic view showing a curved state of the artificial epiglottis shown in FIG. **12(A)**. The dotted line shown in FIG. **13** is a schematic view showing an example of a case in which there are no plurality of portions with different piezoelectric constants as shown in FIG. **12(A)** (i.e., there is one portion with one piezoelectric constant).

[0136] As shown in FIG. **12(A)**, an artificial epiglottis **2011** according to the tenth embodiment is different from the artificial epiglottis **20** according to the first embodiment in that, in the artificial epiglottis **2011**, the piezoelectric element is composed of a plurality of portions with different piezoelectric constants. The basic configuration of the artificial epiglottis **2011** is the same as that of the artificial epiglottis **20**, and descriptions of the same parts will be omitted.

[0137] The artificial epiglottis **2011** includes a first piezoelectric element **211I** and a second piezoelectric element **212I**.

[0138] In this exemplary aspect, the first piezoelectric element **211I** includes a first portion **2111** and a second portion **2112**. The first portion **2111** and the second portion **2112** are arranged in the order of the first portion **2111** and the second portion **2112** from one end (e.g., a fixed end to be fixed to the pharynx) to the other end (e.g., a movable end) of the first piezoelectric element **211I** (i.e., in an L direction in the drawing).

[0139] The piezoelectric constant of the second portion **2112** is larger than that of the first portion **2111**.

[0140] The second piezoelectric element **212I** includes a third portion **2121** and a fourth portion **2122**. The third portion **2121** and the fourth portion **2122** are arranged in the order of the third portion **2121** and the fourth portion **2122** from one end (a fixed end to be fixed to the pharynx) to the other end (a movable end) of the second piezoelectric element **212I** (i.e., in the L direction in the drawing).

[0141] The piezoelectric constant of the fourth portion **2122** is larger than that of the third portion **2121**. The piezoelectric constant of the fourth portion **2122** is the same as that of the second portion **2112**, and the piezoelectric constant of the third portion **2121** is the same as that of the first portion **2111**.

[0142] The first driving electrode **221** overlaps the first portion **2111** and the second portion **2112**. The second driving electrode **222** overlaps the third portion **2121** and the fourth portion **2122**. The third driving electrode **223** is disposed between the first portion **2111** and the third portion **2121** and is also disposed between the second portion **2112** and the fourth portion **2122**.

[0143] Thus, the same voltage is applied to the first laminated portion where the first portion **2111** and the third portion **2121** are laminated, and the second laminated portion where the second portion **2112** and the fourth portion **2122** are laminated.

[0144] The piezoelectric constant of the second laminated portion where the second portion **2112** and the fourth portion **2122** are laminated is larger than the piezoelectric constant of the first laminated portion where the first portion **2111** and the third portion **2121** are laminated.

[0145] Therefore, as shown in FIG. **13**, the second laminated portion is curved more than the first laminated portion.

[0146] Here, when the tracheal entrance and the esophagus are small, as shown by the dotted line in FIG. **13**, the artificial epiglottis that does not have a plurality of piezoelectric constants may block a part of the esophagus or make it difficult to bring the artificial epiglottis into close contact

with the tracheal entrance.

[0147] However, the artificial epiglottis **2011** has its portion on the movable end side curved more than its portion on the fixed end side. This configuration allows the opening area of the esophagus to be secured more reliably and allows the artificial epiglottis **2011** to be brought into close contact with the tracheal entrance more reliably.

[0148] At this time, the curvature of the artificial epiglottis **2011** is configured to be adjusted by adjusting the voltage. Therefore, the artificial epiglottis **2011** secures the opening area of the esophagus more reliably according to the wearer, and the artificial epiglottis **2011** is brought into close contact with the tracheal entrance more reliably.

[0149] As shown in FIG. **12(B)**, an artificial epiglottis **2012** is different from the artificial epiglottis **2011** in the configuration of the driving electrodes. The other configurations of the artificial epiglottis **2012** are the same as those of the artificial epiglottis **2011**, and description of the similar parts will be omitted.

[0150] In this exemplary aspect, the artificial epiglottis **2012** includes a first driving electrode **2211**, a second driving electrode **2221**, a third driving electrode **2231**, a fourth driving electrode **2212**, a fifth driving electrode **2222**, and a sixth driving electrode **2232**.

[0151] The first driving electrode **2211** and the fourth driving electrode **2212** are arranged side by side separated from each other in an L direction of the artificial epiglottis **2012**. The second driving electrode **2222** and the fifth driving electrode **2212** are arranged side by side separated from each other in the L direction of the artificial epiglottis **2012**. The third driving electrode **2231** and the sixth driving electrode **2232** are arranged side by side separated from each other in the L direction of the artificial epiglottis **2012**.

[0152] The first driving electrode **2211** and the third driving electrode **2231** are disposed with the first portion **2111** of the first piezoelectric element **211I** sandwiched therebetween. The fourth driving electrode **2212** and the sixth driving electrode **2232** are disposed with the second portion **2112** of the first piezoelectric element **211I** sandwiched therebetween.

[0153] The second driving electrode **2221** and the third driving electrode **2231** are disposed with the third portion **2121** of the second piezoelectric element **212I** sandwiched therebetween. The fifth driving electrode **2222** and the sixth driving electrode **2232** are disposed with the fourth portion **2122** of the second piezoelectric element **212I** sandwiched therebetween.

[0154] The first driving electrode **2211** and the fourth driving electrode **2212** are connected to a first cable **4112**. The second driving electrode **2221** and the fifth driving electrode **2222** are connected to a second cable **4212**. The third driving electrode **2231** and the sixth driving electrode **2232** are connected to a third cable **4312**.

[0155] With such a configuration, the artificial epiglottis **2012** exhibits the same effects as the artificial epiglottis **2011**.

[0156] As shown in FIG. **12(C)**, an artificial epiglottis **2013** is different from the artificial epiglottis **2012** in the configuration of the cables connected to the driving electrodes. The other configurations of the artificial epiglottis **2013** are the same as those of the artificial epiglottis **2012**, and descriptions of the same parts will be omitted.

[0157] The artificial epiglottis **2013** includes a first cable **411**, a second cable **421**, a third cable **431**, a fourth cable **412**, a fifth cable **422**, and a sixth cable **432**.

[0158] The first cable **411** is connected to the first driving electrode **2211**, and the fourth cable **412** is connected to the fourth driving electrode **2212**. The second cable **421** is connected to the second driving electrode **2221**, and the fifth cable **422** is connected to the fifth driving electrode **2222**. The third cable **431** is connected to the third driving electrode **2231**, and the sixth cable **432** is connected to the sixth driving electrode **2232**.

[0159] With such a configuration, the artificial epiglottis **2013** exhibits the same effects as the artificial epiglottises **2011** and **2012**.

[0160] Further, in the artificial epiglottis **2013**, it is possible to apply different voltages to the

portion on the fixed end side and to the portion on the movable end side. Therefore, in the artificial epiglottis **2013**, the amount of curvature can be further adjusted. Thus, with the artificial epiglottis **2013**, the shape adjustment according to the wearer can be performed more accurately.

[0161] In the present embodiment, a configuration is described in which a plurality of portions having different piezoelectric constants are disposed along the L direction of the artificial epiglottis. However, it is also possible to provide a configuration in which a plurality of portions having different piezoelectric constants are disposed along a W direction (a direction orthogonal to the L direction) of the artificial epiglottis. Further, the number of the portions having different piezoelectric constants is not limited to 2 but may be 3 or more.

Eleventh Exemplary Embodiment

[0162] An artificial epiglottis and a swallowing and breathing switching device according to an eleventh exemplary embodiment will be described with reference to the drawings. FIG. **14** is a view showing a configuration of the swallowing and breathing switching device according to the eleventh exemplary embodiment.

[0163] As shown in FIG. **14**, a swallowing and breathing switching device **10J** according to the eleventh embodiment is different from the swallowing and breathing switching device **10** according to the first embodiment in that the swallowing and breathing switching device **10J** includes a drive signal generating unit **31J**. The other configurations of the swallowing and breathing switching device **10J** are the same as those of the swallowing and breathing switching device **10**, and descriptions of the same parts will be omitted.

[0164] The drive signal generating unit **31J** includes a drive control circuit **390J** and a power supply circuit **311J**. The power supply circuit **311J** includes a variable voltage generator **3161J** and a variable voltage generator **3162J**.

[0165] The drive control circuit **390J** is configured to control, based on the depressed state and the depressed history of the physical switch **39**, the output voltages of the variable voltage generator **3161J** and the variable voltage generator **3162J**, that is, the voltages supplied to the artificial epiglottis **20**.

[0166] For example, when detecting that the physical switch **39** is pressed, the drive control circuit **390J** is configured to perform an output control of a first state. In the output control of the first state, the drive control circuit **390J** performs a control to output +3V to the variable voltage generator **3161J** and output -3V to the variable voltage generator **3162J**.

[0167] Next, when detecting that the physical switch **39** is pressed again, the drive control circuit **390J** is configured to perform an output control of a second state. In the output control of the second state, the drive control circuit **390J** is configured to perform a control to output -3V to the variable voltage generator **3161J** and output +3V to the variable voltage generator **3162J**.

[0168] Next, when detecting that the physical switch **39** is pressed again, the drive control circuit **390J** is configured to perform the output control of the first state. In the output control of the first state, the drive control circuit **390J** is configured to perform a control to output +3V to the variable voltage generator **3161J** and output -3V to the variable voltage generator **3162J**.

[0169] Hereinafter, the drive control circuit **390J** repeats output control of the first state and output control of the second state as described above.

[0170] With such a configuration, the swallowing and breathing switching device **10J** can easily control the voltages supplied to the artificial epiglottis **20**. Further, the swallowing and breathing switching device **10J** can shape the artificial epiglottis **20** as desired by the wearer at a time desired by the wearer.

[0171] In general, it is noted that the configurations of the embodiments described above can be suitably combined, and effects corresponding to the respective combinations can be achieved. For example, the configuration of the swallowing and breathing switching device **10E** according to the sixth embodiment and the configuration of the swallowing and breathing switching device **10F** according to the seventh embodiment can be combined. The swallowing and breathing switching

device obtained by such a combination can switch between a state in which the entrance of the trachea is not blocked and a state in which the entrance of the trachea is blocked by using the brain wave and the myoelectricity. Therefore, aspiration is suppressed more reliably.

[0172] Next, the features of the exemplary embodiments according to the present disclosure will be collectively described.

[0173] In an exemplary aspect <1>, an artificial epiglottis is provided comprising: a membrane-like piezoelectric element configured to bend depending on a voltage; and a driving electrode that applies the voltage to the piezoelectric element. The piezoelectric element is structurally configured to be disposed on a larynx, and the piezoelectric element is configured to become a first shape in which an entrance of a trachea is blocked or a second shape in which the entrance of the trachea is opened, depending on the voltage.

[0174] In an exemplary aspect <2>, for the artificial epiglottis according to exemplary aspect <1>, the piezoelectric element includes a plurality of piezoelectric elements, the plurality of piezoelectric elements are laminated in a state in which membrane surfaces thereof are parallel to each other, and the plurality of piezoelectric elements generate synergized bending depending on the voltage.

[0175] In an exemplary aspect <3>, for the artificial epiglottis according to exemplary aspect <2>, the plurality of piezoelectric elements are composed of a first piezoelectric element and a second piezoelectric element, and the first piezoelectric element and the second piezoelectric element have opposite displacement directions when the voltage is applied.

[0176] In an exemplary aspect <4>, a swallowing and breathing switching device is provided that includes the artificial epiglottis according to any one of exemplary aspects <1> to <3>; and a drive signal generating unit configured to generate a drive signal generating the voltage.

[0177] In an exemplary aspect <5>, for the swallowing and breathing switching device according to exemplary aspect <4>, the drive signal generating unit switches between a first state of the drive signal and a second state of the drive signal, and the artificial epiglottis switches between the first shape or the second shape depending on the first state and the second state.

[0178] In an exemplary aspect <6>, for the swallowing and breathing switching device according to exemplary aspect <5>, the drive signal generating unit includes an electric switch, and switches between the first state and the second state depending on opening and short-circuiting of the electric switch.

[0179] In an exemplary aspect <7>, the swallowing and breathing switching device according to exemplary aspect <6>, further includes a physical switch that controls the electric switch.

[0180] In an exemplary aspect <8>, for the swallowing and breathing switching device according to exemplary aspect <5>, the drive signal generating unit includes an alternating-current power supply, and switches between the first state and the second state depending on whether the voltage from the alternating-current power supply is positive or negative.

[0181] In an exemplary aspect <9>, for the swallowing and

[0182] breathing switching device according to any one of exemplary aspects <4> to <8>, the drive signal generating unit is disposed outside a body on which the artificial epiglottis is worn, and the swallowing and breathing switching device further comprises a cable that connects the drive signal generating unit and the artificial epiglottis.

[0183] In an exemplary aspect <10>, for the swallowing and breathing switching device according to any one of exemplary aspects <4> to <8>, the drive signal generating unit is disposed outside a body on which the artificial epiglottis is worn, and the drive signal generating unit comprises: a power transmission coil disposed outside the body; and a power reception coil disposed inside the body, and connected to the artificial epiglottis.

[0184] In an exemplary aspect <11>, the swallowing and breathing switching device according to exemplary aspect <6>, further comprises: a myoelectric sensor that detects movement of muscles of an oral cavity of a body on which the artificial epiglottis is worn, and generates a myoelectric

detection signal; and a switch control circuit that controls the opening and short-circuiting of the electric switch based on the myoelectric detection signal.

[0185] In an exemplary aspect <12>, the swallowing and breathing switching device according to exemplary aspect <11>, further comprises: a memory that stores a relationship between the myoelectric detection signal and swallowing and breathing; and a switch control circuit that controls the opening and short-circuiting of the electric switch based on the relationship between the myoelectric detection signal and swallowing and breathing stored in the memory.

[0186] In an exemplary aspect <13>, the swallowing and breathing switching device according to exemplary aspect <6>, further comprises: a brain wave sensor that detects a brain wave of a body on which the artificial epiglottis is worn, and generates a brain wave detection signal; and a switch control circuit that controls the opening and short-circuiting of the electric switch based on the brain wave detection signal.

[0187] In an exemplary aspect <14>, the swallowing and breathing switching device according to exemplary aspect <13>, further comprises: a memory that stores a relationship between the brain wave detection signal and swallowing and breathing; and a switch control circuit that controls the opening and short-circuiting of the electric switch based on the relationship between the brain wave detection signal and swallowing and breathing stored in the memory.

[0188] In an exemplary aspect <15>, for the swallowing and breathing switching device according to any one of exemplary aspects <4> to <14>, the drive signal generating unit includes a programmable variable voltage device.

[0189] In an exemplary aspect <16>, the swallowing and breathing switching device according to exemplary aspect <15>, further includes: a memory that stores a relationship between a bending angle of the artificial epiglottis for a wearer of the artificial epiglottis and a voltage corresponding to the bending angle; and a drive control circuit that controls an output voltage of the programmable variable voltage device. In this aspect, the drive control circuit controls the output voltage of the programmable variable voltage device based on the relationship between the bending angle of the artificial epiglottis for the wearer of the artificial epiglottis and the voltage corresponding to the bending angle stored in the memory.

REFERENCE SIGNS LIST

[0190] **10, 10A, 10B, 10C, 10D, 10E, 10F, 10G, 10H, 10J** swallowing and breathing switching device [0191] **20, 20A, 20B, 20I1, 20I2, 20I3** artificial epiglottis [0192] **31, 31A, 31B, 31C, 31D, 31G, 31H** drive signal generating unit [0193] **32** orthosis [0194] **39** physical switch [0195] **40, 40A, 40B** cable [0196] **41** first cable [0197] **42** second cable [0198] **43** third cable [0199] **49** connector [0200] **51** myoelectric sensor [0201] **52** control signal generating unit [0202] **61** brain wave sensor [0203] **62** control signal generating unit [0204] **211, 211I** first piezoelectric element [0205] **2111** first portion [0206] **2122** second portion [0207] **212, 212I** second piezoelectric element [0208] **2121** third portion [0209] **2122** fourth portion [0210] **221, 2211** first driving electrode [0211] **222, 2221** second driving electrode [0212] **223, 2231** third driving electrode [0213] **2212** fourth driving electrode [0214] **2222** fifth driving electrode [0215] **2232** sixth driving electrode [0216] **310** switch control circuit [0217] **311, 311A, 311B, 311C, 311G, 311J** power supply circuit [0218] **312, 313** electric switch [0219] **314** power transmission control unit [0220] **315** power transmission coil [0221] **316** power reception coil [0222] **317** power reception control unit [0223] **318** connector [0224] **390, 390J** drive control circuit [0225] **41, 41I2, 411** first cable [0226] **42, 42I2, 421** second cable [0227] **43, 43I2, 431** third cable [0228] **412** fourth cable [0229] **422** fifth cable [0230] **432** sixth cable [0231] **3161, 3162, 3161J, 3162J** variable voltage generator [0232] **3163** memory [0233] **AC** alternating-current power supply [0234] **C20** cover [0235] **DC1, DC10, DC2** direct-current power supply [0236] **H31** housing

Claims

1. An artificial epiglottis comprising: a membrane-like piezoelectric element configured to bend in response to a voltage applied thereto; and a driving electrode configured to apply the voltage to the piezoelectric element, wherein the piezoelectric element is structurally configured to be disposed on a larynx, and wherein the piezoelectric element is configured to bend, in response to the voltage, to become a first shape to block an entrance of a trachea and to become a second shape to open the entrance of the trachea.
2. The artificial epiglottis according to claim 1, wherein: the piezoelectric element includes a plurality of piezoelectric elements that are laminated in a state in which membrane surfaces thereof are parallel to each other, and the plurality of piezoelectric elements are configured to generate synergized bending depending on the voltage applied by the driving electrode.
3. The artificial epiglottis according to claim 2, wherein the plurality of piezoelectric elements include a first piezoelectric element and a second piezoelectric element that have opposite displacement directions from each other when the voltage is applied thereto.
4. A swallowing and breathing switching device comprising: the artificial epiglottis according to claim 1; and a drive signal generating unit configured to generate a drive signal generating the voltage.
5. The swallowing and breathing switching device according to claim 4, wherein: the drive signal generating unit is configured to switch between a first state of the drive signal and a second state of the drive signal, and the artificial epiglottis is configured to switch between the first shape or the second shape depending on the first state and the second state, respectively.
6. The swallowing and breathing switching device according to claim 5, wherein the drive signal generating unit includes an electric switch that is configured to switch between the first state and the second state depending on opening and short-circuiting of the electric switch.
7. The swallowing and breathing switching device according to claim 6, further comprising a physical switch configured to control the electric switch.
8. The swallowing and breathing switching device according to claim 5, wherein the drive signal generating unit includes an alternating-current power supply, and is configured to switch between the first state and the second state depending on whether the voltage from the alternating-current power supply is positive or negative.
9. The swallowing and breathing switching device according to claim 4, wherein: the drive signal generating unit is disposed outside a body on which the artificial epiglottis is worn, and the swallowing and breathing switching device further comprises a cable that connects the drive signal generating unit to the artificial epiglottis.
10. The swallowing and breathing switching device according to claim 4, wherein: the drive signal generating unit is disposed outside a body on which the artificial epiglottis is worn, and the drive signal generating unit comprises: a power transmission coil disposed outside the body; and a power reception coil disposed inside the body, and connected to the artificial epiglottis.
11. The swallowing and breathing switching device according to claim 6, further comprising: a myoelectric sensor configured to detect a movement of muscles of an oral cavity of a body on which the artificial epiglottis is worn, and to generate a myoelectric detection signal based on the detected movement of muscles; and a switch control circuit configured to control the opening and short-circuiting of the electric switch based on the myoelectric detection signal.
12. The swallowing and breathing switching device according to claim 11, further comprising: a memory configured to store a relationship between the myoelectric detection signal and swallowing and breathing; and a switch control circuit configured to control the opening and short-circuiting of the electric switch based on the relationship between the myoelectric detection signal and swallowing and breathing stored in the memory.

- 13.** The swallowing and breathing switching device according to claim 6, further comprising: a brain wave sensor configured to detect a brain wave of a body on which the artificial epiglottis is worn, and to generate a brain wave detection signal based on the detected brain wave of the body; and a switch control circuit configured to control the opening and short-circuiting of the electric switch based on the brain wave detection signal.
- 14.** The swallowing and breathing switching device according to claim 13, further comprising: a memory configured to store a relationship between the brain wave detection signal and swallowing and breathing; and a switch control circuit configured to control the opening and short-circuiting of the electric switch based on the relationship between the brain wave detection signal and swallowing and breathing stored in the memory.
- 15.** The swallowing and breathing switching device according to claim 4, wherein the drive signal generating unit includes a programmable variable voltage device.
- 16.** The swallowing and breathing switching device according to claim 15, further comprising: a memory configured to store a relationship between a bending angle of the artificial epiglottis for a wearer of the artificial epiglottis and a voltage corresponding to the bending angle; and a drive control circuit configured to control an output voltage of the programmable variable voltage device based on the relationship between the bending angle of the artificial epiglottis for the wearer of the artificial epiglottis and the voltage corresponding to the bending angle stored in the memory.
- 17.** An artificial epiglottis comprising: a plurality of piezoelectric elements that include a first piezoelectric element and a second piezoelectric element disposed in parallel to each other; and a plurality of driving electrodes that include a first driving electrode on a surface of the first piezoelectric element, a second driving electrode on a surface of the second piezoelectric element, and a third driving electrode disposed between the first piezoelectric element and the second piezoelectric element, wherein the plurality of piezoelectric elements are structurally configured to be disposed on a larynx, wherein the plurality of driving electrodes are configured to apply at least one voltage to the plurality of piezoelectric elements to configure the plurality of piezoelectric elements between a first shape to block an entrance of a trachea and a second shape to open the entrance of the trachea, and wherein the plurality of piezoelectric elements are configured to generate synergized bending depending on the at least one voltage applied by the plurality of driving electrodes.
- 18.** The artificial epiglottis according to claim 17, wherein: the first piezoelectric element and the second piezoelectric element comprise a polylactic acid, and the first piezoelectric element and the second piezoelectric element are configured to extend and contract in a direction parallel to the respective surfaces thereof depending on the at least one voltage applied thereto.
- 19.** A swallowing and breathing switching device comprising: the artificial epiglottis according to claim 17; and a drive signal generating unit configured to generate a drive signal generating the voltage.
- 20.** The swallowing and breathing switching device according to claim 19, wherein: the drive signal generating unit is configured to switch between a first state of the drive signal and a second state of the drive signal, and the artificial epiglottis is configured to switch between the first shape or the second shape depending on the first state and the second state, respectively.
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